WHICH LABELLING SCHEME (TARGET_HUMAN OR TARGET_ASV) IS MORE CHALLENGING TO PREDICT FROM HUMAN AND AI MODIFIED AUDIO DATA?

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AICE 2002 - AI and Learning Machines

ABSTRACT

This project investigates whether human perceived labelling (target_human) or AI driven automatic speaker verification (asv) labelling (target_asv) is more difficult to predict from 88 acoustic features for 30 speakers across three different audio systems (Original, X and Y). Care has been taken to avoid data leakage by using speaker-disjoint splits and 5-fold Stratified Group Cross-Validation. Both Random Forest and K-Nearest Neighbours classifiers were used with optimisation. Results show that target_human are significantly more difficult to predict (macro F1 $\tilde{0}$.20–0.28) because of class overlap and the ambiguity that comes from subjective human perception. On pristine audio target_asv are nearly always predicted very deterministically (macro F1 reaching 1.00), however it's performance declines steeply under non-linear system modifications.

1. INTRODUCTION

Mapping of acoustic features to categories has relevance in speech processing and automatic speaker verification (ASV). In this project the predictability of two different coding schemes for audio feature representations is looked at: human perceptual coding (`target_human`) and ASV scoring (`target_asv`).

There are 4,923 samples from 30 speakers (15 male and 15 female) from three different systems: `System_Orig` (original pristine audio), `System_X` (audio processed by AI modification pipeline X), and `System_Y` (audio processed by AI modification pipeline Y). Both individual systems and the full concatenation are examined for generalisation.

2. DATA SPLITS (TASK 1)

To avoid speaker level information leaking (i.e. the model learning features of a particular speaker's vocal tract instead of more generic acoustic features), the anonymised speaker id (sid) was used to perform speaker-disjoint splits using `GroupShuffleSplit`.

2.1. Sizes and Proportions

The dataset comprises 4,923 features across 30 distinct speakers. 24 speakers were allocated($\approx 80\%$, 3,939 samples) to the training/validation pool and 6 speakers were reserved ($\approx 20\%$, 984 samples) strictly for final held-out testing.

2.2. Label Distribution across Splits

Table 1 details the class counts for both labelling schemes across the train/validation and test splits.

Table 1. Label Counts for `target_human` and `target_asv` across Splits

Label Scheme	Class	Train/Val Count	Test Count
`target_human`	goat	872	161
`target_human`	lamb	261	58
`target_human`	sheep	2270	651
`target_human`	wolf	539	111
`target_asv`	goat	507	221
`target_asv`	lamb	154	0
`target_asv`	sheep	2986	760
`target_asv`	wolf	295	0

2.3. Speaker Leakage and Gender Handling

Group splitting guarantees that no speaker present in training/validation sets appear in the test set. Preventing speaker overlap removes optimistic performances. Gender labels (15 male, 15 female) were preserved naturally across splits because speaker IDs are balanced gender-wise. Therefore, 'gender' was dropped as unnecessary metadata.

2.4. Cross-Validation Strategy

Within the training/validation set (24 speakers), `StratifiedGroupKFo` was employed with $K = 5$ folds. This ensures each validation fold contains entirely unseen speakers while preserving class proportions.

3. PRE-PROCESSING (TASK 2)

3.1. Normalisation and Standardisation

All 88 continuous acoustic features were standardised using `StandardScaler` ($\mu = 0, \sigma = 1$). To eliminate data leakage, the scaler was fitted only on training folds during cross-validation and exclusively on the train/validation pool before the final test evaluation.

3.2. Missing Values

Using isnull().sum().sum() verified 0 missing values across all three systems. Therefore, no handling was required.

3.3. Class Balance and Modifications

Both labelling schemes suffer from a high degree of imbalances (for example, `sheep` severely outperforms `lamb`). Instead of using SMOTE or similar synthetic oversampling methods (which can distort the distribution of speech features), the imbalance in the evaluation measures using Balanced Accuracy and Macro F1 was opted. Added noise was not introduced.

4. ALGORITHMS AND PRE-ANALYSIS (TASK 3)

4.1. Algorithm Selection

Two distinct machine learning algorithms were selected:

1. **K-Nearest Neighbors (KNN):** A distance-based classifier suited for localised acoustic similarity which calculates distances from a test datapoint, to all of the other datapoints in the dataset

2. **Random Forest (RF):** An ensemble method capable of capturing non-linear interactions and handling collinear audio features.

4.2. Pre-choice Analysis

During feature matrix inspection, 88 continuous features with features spanning a wide range of dynamic were found. This motivated the use of feature scaling to improve performance in KNN (distance calculations) and RF (split implementations).

5. HYPERPARAMETER TUNING (TASK 6)

Hyperparameters were optimised via 5-fold Stratified Group Cross-Validation on training/validation data.

- **KNN:** $n_neighbors \in \{3, 5, 7, 11\}$ and $weights \in \{\text{'uniform'}, \text{'distance'}\}$.
- **Random Forest:** $n_estimators \in \{50, 100, 200\}$ and $max_depth \in \{\text{None}, 10, 20\}$.

The optimal configurations selected based on mean validation Macro F1 were used for the final model evaluation.

6. RESULTS ACROSS SYSTEMS (TASK 4) & EVALUATION METRICS (TASK 5)

6.1. Evaluation Metrics

Models were evaluated using: (1) **Accuracy**, (2) **Macro F1-Score** (unweighted mean across classes to penalise poor minority performance), (3) **Weighted F1-Score**, and (4) **Balanced Accuracy** (mean recall across classes).

6.2. Results Table

Table 2 provides the diversified evaluation across systems, algorithms, and labeling schemes.

6.3. System Differences and Disagreements

There are large disparities observed between accuracy and Macro F1 on imbalanced systems. For example the ASV System_Original KNN has 97.% accuracy yet only 0.3287 Macro F1 because there are no test samples for some of the minority classes such as `lamb` and `wolf`.

For System_Original the ASV labels appear deterministic and can be easily predicted (RF has 1.0000 Macro F1). This is expected because the algorithm makes deterministic predictions. In System_Y the ASV performance drops significantly to 45.57%. However, the Human performance is relatively maintained with a Macro F1 value of 0.4000. This is due to the average person's ability to abstract high level phonetic features despite low level audio transformations.

7. FINDINGS AND CONCLUSIONS (TASK 7)

7.1. Answer to Research Question

`target_human` is substantially more challenging to predict than `target_asv`. As shown in the Task 4 Results Table, across the Combined system, Random Forest achieved a Macro-F1 of only 0.202 and KNN achieved 0.252, with balanced accuracies hovering near random guessing ($\approx$ 0.21–0.25). Models struggled equally across all four classes because acoustic representations alone cannot establish clear decision boundaries for labels shaped by subjective human interpretation. Human perception involves ambiguity and subjectivity to the listener in terms of psychoacoustical factors that are tied to perception. However ASV involves deterministic scoring without ambiguity introduced by human perception on pristine signals.

Table 2. Classification Performance Across Audio Systems and Label Schemes

System	Label Scheme	Algorithm	Accuracy	Macro F1	Weighted F1	Balanced Acc
System_Original	Human	KNN	0.3578	0.1532	0.3936	0.2752
System_Original	Human	Random Forest	0.3853	0.1403	0.3707	0.2917
System_X	Human	KNN	0.4220	0.2602	0.4295	0.3331
System_X	Human	Random Forest	0.3731	0.2282	0.3778	0.2955
System_Y	Human	KNN	0.7370	0.2829	0.7085	0.4414
System_Y	Human	Random Forest	0.6667	0.4000	0.6679	0.3993
Combined	Human	KNN	0.5127	0.2523	0.5025	0.2583
Combined	Human	Random Forest	0.5036	0.2023	0.4649	0.2189
System_Original	ASV	KNN	0.9725	0.3287	0.9860	0.9725
System_Original	ASV	Random Forest	1.0000	1.0000	1.0000	1.0000
System_X	ASV	KNN	0.8226	0.2257	0.9027	0.8226
System_X	ASV	Random Forest	0.8746	0.3110	0.9331	0.8746
System_Y	ASV	KNN	0.3180	0.1909	0.2295	0.4537
System_Y	ASV	Random Forest	0.4557	0.3053	0.4320	0.5727
Combined	ASV	KNN	0.7136	0.2350	0.6688	0.4846
Combined	ASV	Random Forest	0.7930	0.3005	0.7544	0.5873

7.2. Limitations

Key limitations include the sample size constraint of 30 speakers, absence of minority classes in certain test subsets due to speaker-disjoint splits, and reliance on insubstantial classifiers.

7.3. Future Work

A future study that I would undertake would dissect deeper learning architectures trained across an expanded multi-emphasis body with continuous soft human perception targets rather than discrete hard labels.

8. REFERENCES

[1] Python Core Team. (2026). *The Python Standard Library: Complete Global Namespace Import*. Python Software Foundation. Available: `https://docs.python.org/3/library/`